

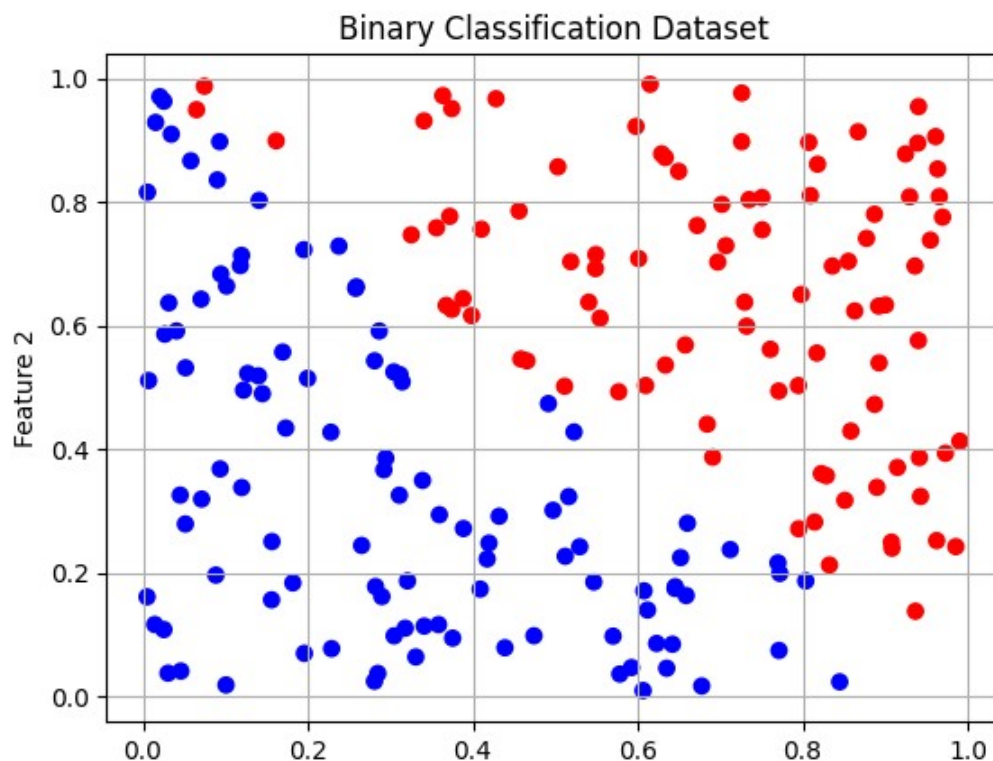
✓ Day 27: Linear Binary Classification in TensorFlow

```
import numpy as np
import tensorflow as tf
import matplotlib.pyplot as plt

# Generating synthetic binary classification data
np.random.seed(42)
X_train = np.random.rand(200, 2) # 200 samples, 2 features
y_train = (X_train[:, 0] + X_train[:, 1] > 1).astype(int) # Binary labels (0 or 1)
import numpy as np
import tensorflow as tf
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# Visualizing the data
plt.scatter(X_train[:, 0], X_train[:, 1], c=y_train, cmap='bwr')
plt.title('Binary Classification Dataset')
plt.xlabel('Feature 1')
plt.ylabel('Feature 2')
plt.grid(True)
plt.show()
```



Feature 1

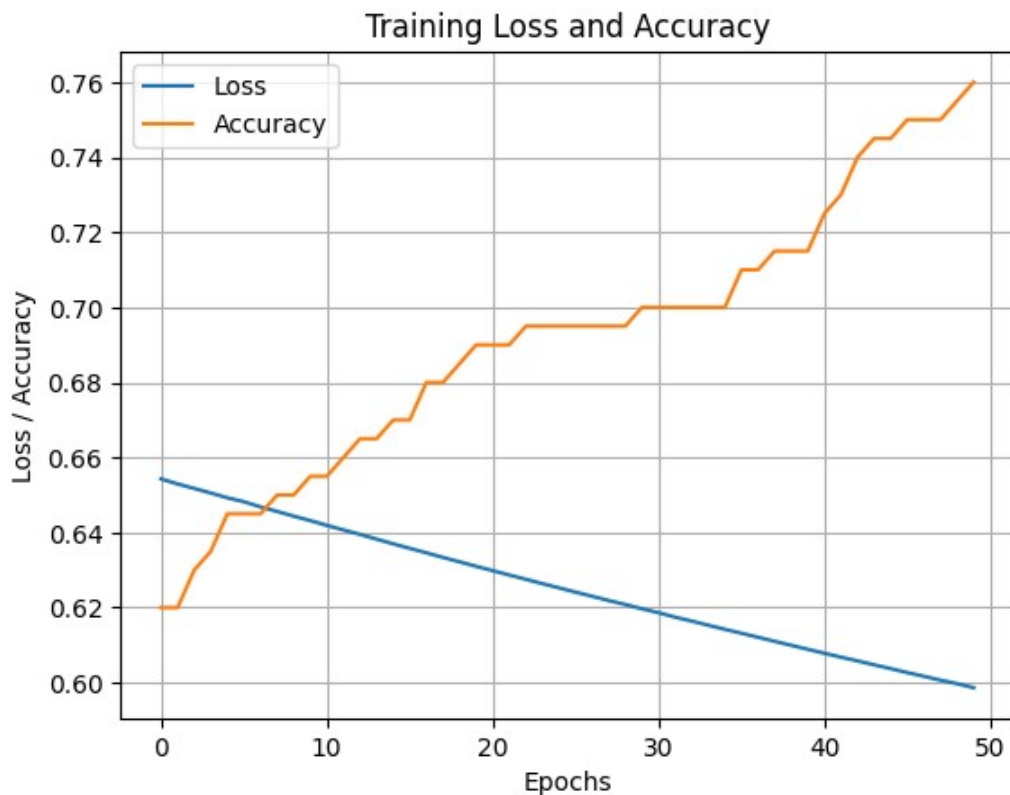
```
# Building a simple linear model
model = tf.keras.Sequential([
    tf.keras.layers.Dense(1, activation='sigmoid', input_shape=(2,)) # Sigmoid activation for bi
])

# Compiling the model with Binary Crossentropy loss and Adam optimizer
model.compile(optimizer='adam', loss='binary_crossentropy', metrics=['accuracy'])
```

```
⚡ /usr/local/lib/python3.10/dist-packages/keras/src/layers/core/dense.py:87: UserWarning: Do r
    super().__init__(activity_regularizer=activity_regularizer, **kwargs)
```

```
# Training the model
history = model.fit(X_train, y_train, epochs=50, batch_size=16, verbose=0)
```

```
# Plotting the training loss and accuracy
plt.plot(history.history['loss'], label='Loss')
plt.plot(history.history['accuracy'], label='Accuracy')
plt.title('Training Loss and Accuracy')
plt.xlabel('Epochs')
plt.ylabel('Loss / Accuracy')
plt.legend()
plt.grid(True)
plt.show()
```



```
# Generating test data
X_test = np.random.rand(50, 2)
y_test = (X_test[:, 0] + X_test[:, 1] > 1).astype(int)

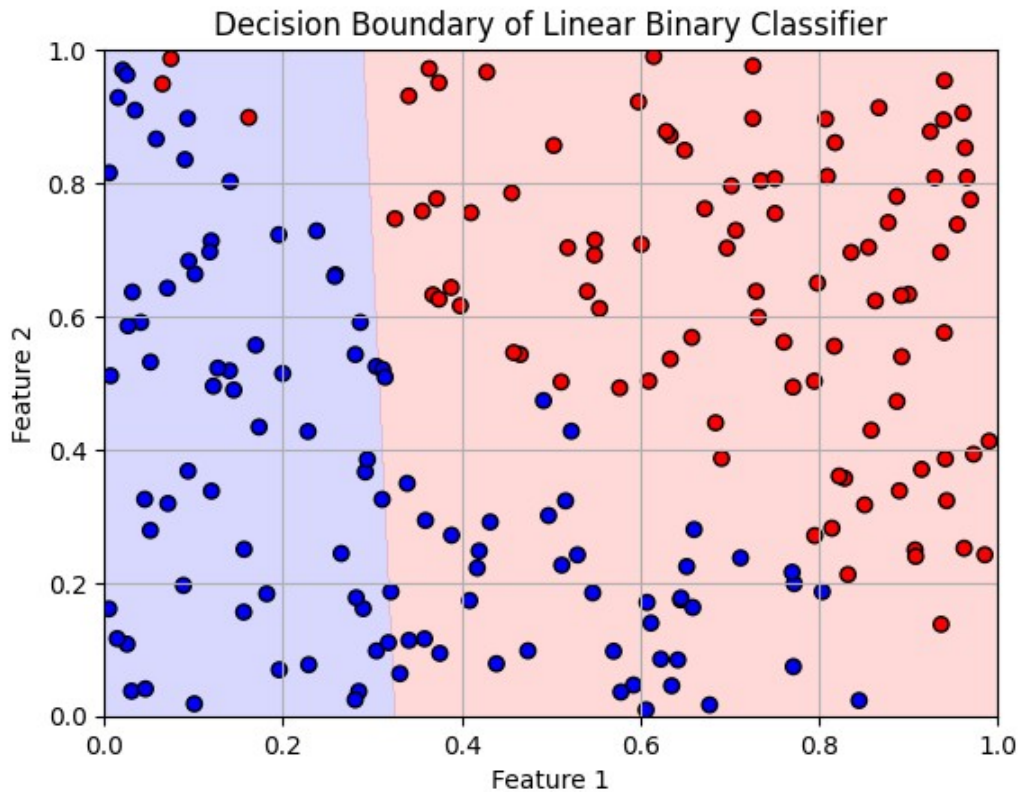
# Evaluating the model on test data
test_loss, test_accuracy = model.evaluate(X_test, y_test)
print(f'Test Loss: {test_loss:.4f}, Test Accuracy: {test_accuracy:.4f}')
```

2/2 ————— 0s 14ms/step - accuracy: 0.7329 - loss: 0.6100
Test Loss: 0.6035, Test Accuracy: 0.7400

```
# Visualizing the decision boundary
x1, x2 = np.meshgrid(np.linspace(0, 1, 100), np.linspace(0, 1, 100))
grid = np.c_[x1.ravel(), x2.ravel()]
preds = model.predict(grid).reshape(x1.shape)

plt.contourf(x1, x2, preds, levels=[0, 0.5, 1], cmap='bwr', alpha=0.3)
plt.scatter(X_train[:, 0], X_train[:, 1], c=y_train, cmap='bwr', edgecolor='k')
plt.title('Decision Boundary of Linear Binary Classifier')
plt.xlabel('Feature 1')
plt.ylabel('Feature 2')
plt.grid(True)
plt.show()
```

313/313 ————— 0s 1ms/step



Linear binary classification is a fundamental technique for many machine learning tasks. This model uses a simple linear decision boundary to classify data into two categories, showing how TensorFlow can be used to build and visualize powerful models quickly. """

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